

ProspectMap: The Two Ways a Mineral Prospectivity Map Lies, Quantified on the Real US Mississippi-Valley-Type Belt

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 Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_ProspectMap

Abstract. A mineral prospectivity map stacks geophysical, geochemical and structural evidence over a study area and outputs, per cell, the probability that a deposit is there. Such maps are typically presented with an area-under-curve (AUC) and a posterior, and both numbers can be inflated in two specific, well-understood ways. ProspectMap is an interactive tool that computes such a map by Weights of Evidence, a Bayesian log-odds update, and makes those two inflations *first-class*, measuring them rather than hiding them. This report quantifies both on the real US Midcontinent Mississippi-Valley-type zinc-lead belt (the open Critical Minerals Mapping Initiative dataset of Lawley *et al.*, 2022) and on ten synthetic terranes. The first inflation is *evaluation*: reporting AUC under random cross-validation, which lets spatially adjacent cells leak between train and test, overstates prospective skill; on the real belt the Weights-of-Evidence AUC, scored over all map cells, falls from 0.72 under random cross-validation to 0.64 under spatial block cross-validation (an inflation of 0.09, the largest of any case), and a learned classifier, scored on a labelled sample of deposit cells and distance-buffered non-deposit cells, drops from 0.98 to 0.95. The spatial estimate is the one to report; the two models' spatial-CV AUCs come from different cell sets and are not a like-for-like comparison. The second inflation is *modelling*: Weights of Evidence assumes the evidence layers are conditionally independent given a deposit, and it reports large contrasts that look decisive, but on the real belt those contrasts are statistically *insignificant*, every layer's studentized contrast is below the 95% threshold, so the confident-looking weights do not survive their own error bars. The tool displays both corrected diagnostics, the spatial-CV AUC and the studentized significance, beside the inflated values; every number and figure can be regenerated from the released artifacts.

Index Terms. Mineral prospectivity, Weights of Evidence, conditional independence, spatial cross-validation, AUC inflation, Mississippi-Valley-type, critical minerals, model evaluation, reproducibility.

I. INTRODUCTION

WHERE is the next ore deposit most likely to be? Mineral prospectivity mapping answers it by overlaying evidence layers, a magnetic high here, a geochemical anomaly

there, proximity to a favourable structure, over a study-area grid and computing, per cell, a posterior probability of a deposit. The classical method is Weights of Evidence [2], [1]: each binary evidence pattern gets a positive weight W^+ where it is present and W^- where it is absent, their difference is the contrast $C = W^+ - W^-$, and the posterior log-odds of a deposit in a cell is the prior log-odds plus the sum of the present-or-absent weights. The output is a map, and the map is usually reported with two numbers: a high area-under-curve showing it separates known deposits from barren ground, and confident-looking weights. Both are, in a well-documented way, prone to inflation.

ProspectMap is a tool built to expose the inflation. It computes the Weights-of-Evidence posterior live, and it makes first-class the two ways the numbers lie. The first is that an AUC reported under *random* cross-validation is optimistic, because a prospectivity grid is spatially autocorrelated and a random split leaves near-neighbour cells in both train and test, so the model is partly tested on what it has seen; *spatial* cross-validation, holding out contiguous blocks, gives a realistic estimate [5]. The second is that Weights of Evidence assumes the evidence layers are conditionally independent given a deposit, and when they are not the posterior is overconfident [3]; and even when the contrasts look large, they may not be statistically significant. This report quantifies both, on the real US Mississippi-Valley-type zinc-lead belt from the open Critical Minerals Mapping Initiative dataset [4] and on ten synthetic terranes, and reports a modern learned classifier [6] under the same random-versus-spatial contrast. The contribution is the measurement and the display of the two inflations, not a new prospectivity algorithm.

II. THE FIRST LIE: CROSS-VALIDATION INFLATION

The first inflation is a matter of how the map is scored, and it is quantified in Fig. 1a. For every case, the Weights-of-Evidence posterior is evaluated two ways: random five-fold cross-validation, which splits the grid cells at random, and spatial five-fold cross-validation, which holds out contiguous 20×20 -cell blocks so that the test set is geographically separate from the training set; in both, the weights are refitted on

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Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ProspectMap; the tool is available at <https://prospectmap.fasl-work.com>.

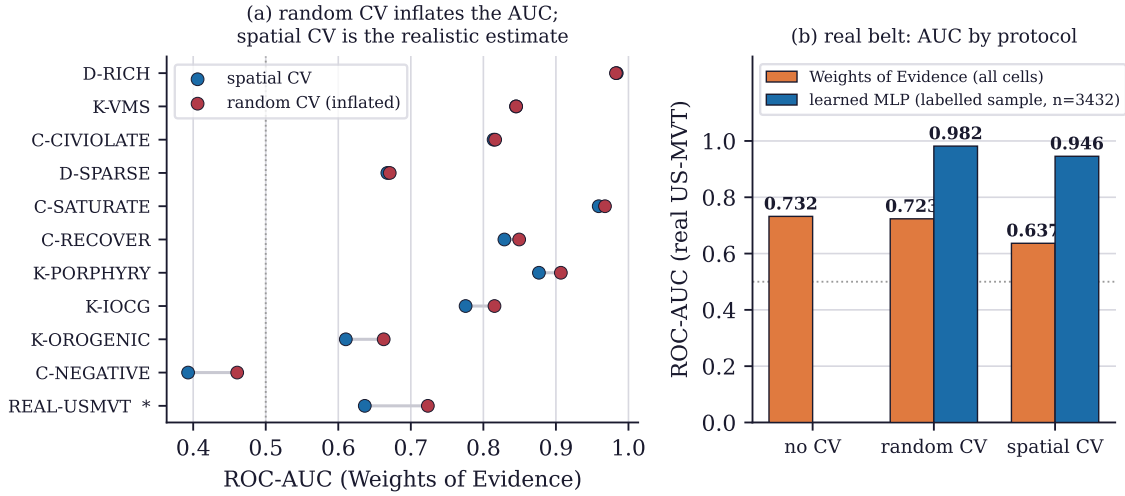


Fig. 1. The evaluation inflation. (a) Weights-of-Evidence AUC under random versus spatial cross-validation, per case (five folds; spatial folds are 20×20 -cell blocks; every map cell scored): random cross-validation (red) is at least as high as spatial (blue) in every case except D-RICH, where spatial exceeds random by 0.001, because adjacent cells leak; the gap is largest on the real US-MVT belt (marked *), where the AUC falls from 0.72 to 0.64. Spatial cross-validation gives the realistic estimate. (b) The real belt, AUC by protocol: Weights of Evidence scored over all map cells without cross-validation (0.732, weights fitted on all deposits), under random (0.723) and under spatial (0.637) cross-validation; the learned classifier scored on its labelled sample (the 858 deposit cells and 2574 sampled non-deposit cells more than two cells from any deposit) under random (0.982) and spatial (0.946) cross-validation. The two models are scored on different cell sets, so their AUCs are not directly comparable.

the training folds' deposits and every map cell receives a held-out score. Random cross-validation is at least as optimistic in every case but D-RICH, where the spatial AUC is higher by 0.001, because a prospectivity grid is strongly spatially autocorrelated and a random split leaves a test cell's near neighbours in the training set, so the model is scored partly on cells like ones it has seen. The gap is real and, on the *real* US Mississippi-Valley-type belt, it is the largest of any case: the Weights-of-Evidence AUC falls from 0.723 under random cross-validation to 0.637 under spatial, an inflation of 0.087. A learned classifier, a small multilayer perceptron on the six continuous evidence layers (Weights of Evidence uses their binarized patterns), cross-validated with the same block size and fold count but scored on a labelled sample of the 858 deposit cells and 2574 sampled non-deposit cells more than two cells from any deposit, falls from 0.982 to 0.946. The spatial estimate is the relevant one; the random one overstates how well the map would find a deposit in unseen ground. Fig. 1b sets these values beside the Weights-of-Evidence AUC without cross-validation (0.732, weights fitted and scored on the whole map). The spatial-CV AUCs of the learned classifier (0.946) and of Weights of Evidence (0.637) are computed on different cell sets and averaged differently (the mean of per-fold AUCs for the classifier, pooled held-out scores for Weights of Evidence), so their difference is not a like-for-like comparison of the two methods.

III. THE SECOND LIE: OVERCONFIDENT WEIGHTS

The second inflation is in the weights themselves (Fig. 2). Weights of Evidence assumes the evidence layers are conditionally independent given a deposit; when correlated layers are stacked, their weights double-count the same signal and the posterior becomes overconfident, which is why a conditional-

independence test [3] belongs in any Weights-of-Evidence workflow, and ProspectMap runs one. But there is a simpler and equally important check that is often skipped: whether a contrast is statistically significant at all. On the real US-MVT belt the six evidence layers have large raw contrasts, up to about 6, which a practitioner reading only the contrast would take as strong evidence. Their studentized contrasts $C/s(C)$, however, are all between 0.2 and 0.9, well below the 1.645 threshold for 95% significance: with sparse deposits and noisy layers the standard error of each weight is large, and none of the contrasts clears its own error bar. The map built from them may still be useful for ranking, but presenting its weights as decisive evidence is the second way the map lies, and showing the studentized contrast beside the raw one is the correction.

IV. DISCUSSION AND SCOPE

The two inflations are independent and both matter. A map can have significant weights and still report an inflated AUC because it was scored with random cross-validation; or it can have a spatially cross-validated AUC and still oversell weights that are not significant. ProspectMap's contribution is to compute the Weights-of-Evidence map and then put the deflating diagnostics next to it, the spatial-CV AUC beside the random one, the studentized contrast beside the raw one, and a conditional-independence test, so a user sees the corrected diagnostics beside the inflated ones, and a learned classifier is reported under the same random-versus-spatial contrast (0.982 against 0.946 on its labelled sample). The ten terrane cases are synthetic, used to exercise the diagnostics under known ground truth; the one real case is the open Critical Minerals Mapping Initiative US-MVT dataset [4], a real belt, on which spatial cross-validation removes the largest AUC inflation of any case and no evidence layer's contrast is significant. The tool is a

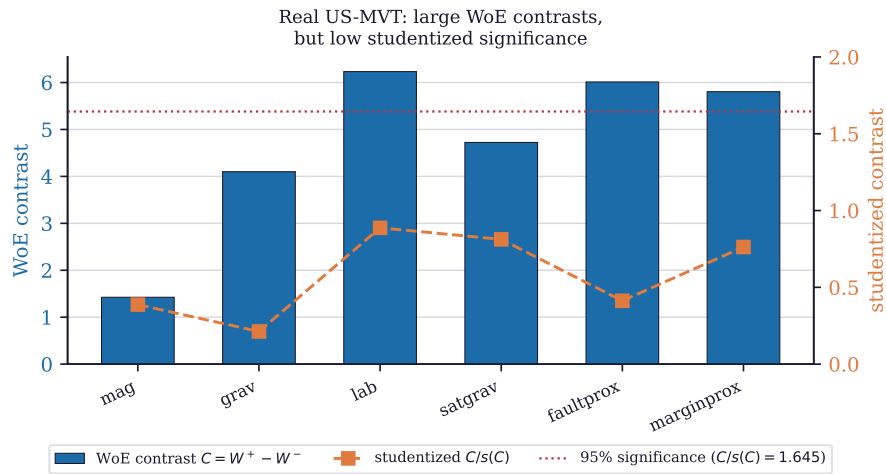


Fig. 2. The modelling inflation. Weights-of-Evidence contrasts $C = W^+ - W^-$ (bars) and their studentized significance $C/s(C)$ (line) for the six evidence layers of the real US-MVT belt. The raw contrasts are large (up to 6), which looks decisive, but every layer’s studentized contrast is below the 95% significance threshold of 1.645: on real, sparse deposit data the weights carry large standard errors and are not statistically significant, so the confident-looking map rests on evidence that does not survive its own error bars.

decision-support and teaching instrument, not an exploration recommendation for any specific ground; it says where the evidence points and, more importantly, how much to trust it. The interface shows the same diagnostics on every map.

V. CONCLUSION

A mineral prospectivity map lies in two well-understood ways: it reports an AUC inflated by random cross-validation, and it presents Weights-of-Evidence contrasts that look decisive but may violate conditional independence or fail their own significance test. ProspectMap measures both. On the real US Mississippi-Valley-type belt the Weights-of-Evidence AUC falls from 0.72 to 0.64 when scored by spatial cross-validation (the largest inflation of any case), a learned classifier scored on a labelled sample falls from 0.98 to 0.95 under the same switch from random to spatial cross-validation, and every evidence layer’s contrast is statistically insignificant despite looking large. The contribution is a reproducible tool that displays the corrected diagnostics, spatial-CV AUC and studentized significance, next to the inflated ones.

APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry each case’s Weights-of-Evidence layers (contrasts and studentized contrasts), posterior summary, pairwise conditional-independence tests, and both random- and

spatial-cross-validation AUC (`case-results.json`), together with the learned classifier’s random- and spatial-CV AUCs on the real belt (`pm-learned-real.json`, whose field `spatial_cv.wofe_roc_auc` holds the Weights-of-Evidence AUC without cross-validation) and the real US-MVT trace (REAL-USMVT, with the Lawley *et al.* provenance). The figure scripts under `manuscripts/prospectivity/figures/` read those artifacts; the pipeline is seeded, so re-running reproduces Fig. 1 and Fig. 2.

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